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comparatively lower priced, flexible, has a longer lifetime and allows continuous update of features and algorithms with the help of software toolsets, such as Altera video and image processing (VIP) suite. This can be used in positron emission tomography (PET), optical coherence tomography (OCT), computerised tomography (CT), X-ray imaging, ultrasound scanning, image guided therapy and so on.

For medical equipment manufacturers. A handheld hardware platform that lets you drag and drop sensor algorithms between the processor and FPGA fabric, and instantly observes resulting trade-offs in performance and power consumption, is being developed by Netherlands based Topic Embedded Products in collaboration with Xilinx. This would serve as a complete clinical reference design platform for medical equipment manufacturers.

In automotive industry. With autonomous driving becoming a reality in coming years, demand for technologically-advanced features, like automated braking, lane tracking, pedestrian detection, traffic sign recognition and parking assistance, have translated into a myriad list of challenges for system designers. With evolving safety, security and reliability standards for a vehicle, FPGAs offer a scalable, highly-reliable and power-friendly platform with reduced form factor and bill of material (BoM). A typical intelligent car has greater than 60 electronic control units (ECUs).

"Designing such a hard, real-time, low-latency gateway unit requires the right combination of hardware and software intellectual properties (IPs), driven by a system software capable of exploiting the best performance from underlying hardware," says Bhaskar Das of Robert Bosch Engineering and Business Solutions. "The flexibility and scalability offered by FPGAs allow system designers to experiment, benchmark and quickly arrive at an optimal system design that rightly balances hardware and software needs," he adds.

Which programmable chip is best suited for an application

Here are a few factors based on which the programmable IC for any application could be selected.

Logic requirement. Logic requirement of an application plays an important role in FPGA selection. If the application is simple and requires only about thousands of logic cells, CPLDs can be used. For low- and medium-complexity projects that require a hundred thousand logic cells or more, low-end and mid-range FPGAs are the options. High-end FPGAs with millions of logic cells are used for very complex applications.

System blocks/design requirement. From simple logic cells, programmable ICs have reached a level of high integration where several features and functions, like DSP blocks, PCIe blocks, processors, mixed signals and memory,

are integrated into the chip. Altera's Arria 10 FPGAs and SoCs have integrated ARM processors, Cypress PSoC integrates configurable analogue and digital peripheral functions, memory and microcontroller on a single chip, and Lattice iCE40 Ultra integrates emerging infra-red remote, barcode, touch, user identification and pedometer functions.

I/O. Input/output requirements, like voltage, SERDES interface standards and others, can be deciding factors while selecting a programmable chip. For example, Altera Arria 10 supports double data rate random access memory (DDR4 RAM) interface with high data rates. Similarly, Xilinx Ultrascale family has a hybrid memory cube (HMC) interface.

Price. The cost of CPLDs is less as compared to FPGAs. Many medium complex designs are more cost-effective at 28nm node of FPGAs. 16nm node is generally for high-end designs and costs more.

What the future holds

Adoption of 14nm FinFET technology has given the programmable industry a much needed boost in terms of high performance, lower cost and lower power, and has bridged the gap between FPGAs and ASICs even closer. Chip manufacturers are now moving towards 10nm technology. Natarajan believes that Intel's partnership in the programmable space will lead to more breakthrough innovations and allow programmable devices to meet the ever-increasing demand of customers.

Gabriel states that the two emerging trends in the industry—data acceleration and network enrichment—are addressing the need for faster and predictable real-time performance, low-power Wi-Fi and Ethernet connections, and rapidly-evolving device security. "As the industry is struggling to make the IoT work, these two trends play directly into the strengths of, while creating great opportunities for, programmable logic," adds Gabriel. ●